

Appendix: Using Hierarchical Models to Estimate Heterogeneous Effects

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1 Replication Analysis Details

To produce Figure 1, I downloaded every replication archive on the Harvard Dataverse for papers in the *American Political Science Review*, *American Journal of Political Science*, and *Journal of Politics*, yielding 1,830 archives. Coverage runs from 2012 to 2024; deposits for 2025 and 2026 remain incomplete. Each paper is dated by its year of publication, taken from the Dataverse citation string, rather than by the year its archive was deposited. The distinction matters more than it appears: journals adopted deposit mandates around 2015, so older articles were archived retroactively, and dating by deposit places only one paper in 2012 and 274 in 2015 against true counts of 17 and 120.

Within each archive I read every file of author-written analysis code, meaning scripts in R, Stata, Python, Stan, and JAGS. This excludes `.ado` files, which are third-party Stata packages rather than the author’s analysis; their help text contains the same syntax the analysis looks for. The 1,830 archives contain 12,179 such files.

1.1 Classification

Rather than match patterns, I gave each file in full to an open-weight language model and asked whether the analysis estimates an effect that varies across units, subgroups, or contexts. The model returns a structured judgment, the methods it identifies, and verbatim lines of code as evidence. Files longer than the context window are split into overlapping chunks and judged separately; a paper counts as estimating heterogeneous effects if any of its files does.

Pattern matching was inadequate for this task in a way worth stating, because it bears on the trend rather than only the level. Regular expressions key on the syntax of an interaction, and that syntax changed over the period. Stata’s factor-variable notation arrived in version 11 and displaced hand-constructed product variables only gradually, so a script from 2012 is far more likely to write `gen treatXpost = treat*post` and enter the result as an ordinary regressor. A pattern-matching classifier misses those, and misses them disproportionately in early years. Scoring the same corpus both ways, the two approaches agree almost exactly on recent papers and diverge sharply on old ones: over 2020 to 2024 both flag 67% of papers, while over 2012 to 2015 pattern matching flags 35% against the classifier’s 58%. Most of the upward trend a pattern-matching measure reports is therefore a change in how interactions are written rather than in how often they are estimated.

I use `openai/gpt-oss-120b`, served through DeepInfra at temperature zero. The prompt, the full text of every request, and every raw model response are archived in the replication materials, so the classification can be verified without re-running inference, and the weights are public so it can be re-run.

1.2 Validation

I hand-adjudicated a stratified random sample of 100 file-chunks, drawn blind: the sample was shuffled and the model’s judgment hidden until each item was answered. Each item

displayed the relevant code with four lines of context on either side. Weighting back to the stratum populations, precision for the measure reported in Figure 1 is above 90%.

Two limits deserve statement. First, recall is not identified by this design. Estimating it requires knowing how often the classifier misses a heterogeneous effect entirely, and over half the corpus falls in the stratum where nothing was flagged and there is consequently nothing to check without reading whole files. Figure 1 is therefore a lower bound. Second, I do not count subgroup analysis conducted by splitting the sample and fitting the same model separately, though that estimator is algebraically identical to a fully crossed interaction for the group-specific point estimates. Counting it would raise the 2024 share from 65% to roughly 86%. I exclude it because it is the least reliably measured category: agreement between two independently prompted models on subgroup splitting was $\kappa = 0.49$, against 0.92 for the measure reported here.

2 Simulation Data Generating Process

This section summarizes the data generating processes for the simulations in the main text. All of them share one structure and differ only in two places: the probabilities used to assign the grouping variables, which control how uneven the groups are, and the distribution from which the true group treatment effects are drawn. Every condition fixes the sample size at $n = 2000$ and varies the number of groups.

2.1 Common Structure

I generate k binary grouping variables, where $k \in \{3, 4, 5, 6, 7\}$:

$$x_{ij} \sim \text{Bernoulli}(p_j) \quad \text{for } j = 1, \dots, k$$

The simulation defines the groups by unique combinations of these variables, producing 2^k groups. With three modifiers, there are 8 groups, while 7 modifiers produces 128 groups. Most processes set $p_j = 0.5$ for all j , which makes the groups equal in expectation. The two unbalanced processes instead space the probabilities evenly over $[0.35, 0.65]$; with $k = 7$ this is already severe, giving expected cell sizes that run from roughly one observation to roughly one hundred.

The treatment indicator is randomly assigned, also with a bernoulli draw:

$$T_i \sim \text{Bernoulli}(0.5)$$

Three control variables predict the baseline outcome but do not modify the treatment

effect:

$$\begin{aligned} z_{i1} &\sim N(0, 1) \\ z_{i2} &\sim \text{Bernoulli}(0.5) \\ z_{i3} &\sim N(0, 1) \end{aligned}$$

The outcome is then generated as:

$$\begin{aligned} y_i &= \beta_1 z_{i1} + \beta_2 z_{i2} + \beta_3 z_{i3} + \tau_{g[i]} T_i + \epsilon_i \\ \epsilon_i &\sim N(0, \sigma_y) \quad \text{where } \sigma_y = 1 \end{aligned}$$

I also fix the control coefficients $\beta = (0.4, -0.3, 0.25)$. Throughout, $\mu_\tau = 0.2$ and $\sigma_\tau = 0.3$, and each process is scaled so that the marginal standard deviation of the group effects stays near σ_τ . Holding that constant keeps root mean squared error comparable across processes.

2.2 The Eight Data Generating Processes

Table ?? lists them. Let $\mathbf{Z}_g$ collect the modifier values for group g , and let

$$\gamma = (0.20, -0.15, 0.12, -0.10, 0.08, -0.06, 0.05)$$

truncated to the first k entries. Where a systematic component $\mathbf{Z}_g \gamma^*$ appears it is centered and rescaled to the standard deviation shown.

The last process deserves comment, because it is the one designed against the hierarchical model. In every other process the group effects contain an idiosyncratic component that an additive interaction model cannot represent by construction, which is a comparison the hierarchical model cannot lose. Under the exactly additive process the additive model is correctly specified, and the hierarchical model is paying to estimate a group variance that is genuinely zero. It does lose there, by roughly three percent of root mean squared error. That is the scope condition: partial pooling costs little when it is unnecessary and gains substantially when it is not.

2.3 Replication and Estimation

Each condition uses 200 Monte Carlo replications, and the shaded bands in the figures give 95% Monte Carlo intervals computed from the spread of per-replication root mean squared error. Seeds are a deterministic function of the process, the number of modifiers, and the replication index, so any single condition can be reproduced in isolation.

Accuracy and errors of inference come from different estimators, for a reason worth stating. Root mean squared error requires no standard error, so it is computed over all 200 replications using restricted maximum likelihood, which is fast enough to make that many replications affordable. Restricted maximum likelihood corresponds to no prior at all and therefore regularizes less than any half- t or half-normal prior, so it is a conservative stand-in: where the

Process	p_j	Group treatment effects
Normal	0.5	$\tau_g \sim N(\mu_\tau, \sigma_\tau)$
Systematic + deviation	0.5	$\tau_g = \mu_\tau + \mathbf{Z}_g \boldsymbol{\gamma}^* + \nu_g$, with $\mathbf{Z}_g \boldsymbol{\gamma}^*$ scaled to SD $\sigma_\tau/\sqrt{2}$ and $\nu_g \sim N(0, \sigma_\tau/\sqrt{2})$
Sparse	0.5	$\tau_g = \mu_\tau + d_g \nu_g$, where $d_g \sim \text{Bernoulli}(0.15)$ and $\nu_g \sim N(0, \sigma_\tau/\sqrt{0.15})$: most groups share one common effect and a minority genuinely differ
Bimodal	0.5	$\tau_g = \mu_\tau \pm 0.3 + \nu_g$, sign drawn Bernoulli(0.5), $\nu_g \sim N(0, 0.1)$: two clusters, with little mass at the grand mean
Student- t	0.5	$\tau_g = \mu_\tau + \sigma_\tau t_3/\sqrt{3}$: heavy tails, so genuine outlier groups exist
Unequal group sizes	[0.35, 0.65]	$\tau_g \sim N(\mu_\tau, \sigma_\tau)$: effects as in the normal case, but groups differ enormously in size
Unequal sizes, effects correlated with size	[0.35, 0.65]	$\tau_g = \mu_\tau + \sigma_\tau(0.8z_g + \sqrt{1 - 0.8^2} \nu_g)$, where z_g is the standardized log expected size of group g : the groups with the most data pull the grand mean toward themselves
Exactly additive	0.5	$\tau_g = \mu_\tau + \mathbf{Z}_g \boldsymbol{\gamma}^*$ scaled to SD σ_τ , with <i>no</i> idiosyncratic component

Table 1. Data generating processes. All share the common structure above and differ only in the assignment probabilities and the distribution of the group treatment effects.

additive model loses to it, the additive model loses to the Bayesian model by more. Sign and exaggeration errors, however, depend on which estimates are declared distinguishable from zero, and the mixed model has no convenient interval for a group treatment effect. The prediction error is $\mathbf{x}'_g(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) + (\hat{\theta}_g - \theta_g)$, and the covariance between those two terms is strongly negative, because the varying slope absorbs error in the fixed part. Adding the two variances yields intervals covering 0.99 against a nominal 0.95, while the conditional variance alone covers 0.84. Those quantities are therefore computed from posterior intervals, whose coverage across these processes measures between 0.94 and 0.97.

3 Tomz and Weeks Reanalysis

This section summarizes the hierarchical models of ? and details the underlying R code. Both models use the nonlinear lambda specification in brms, which separates the outcome into two linked equations: a heterogeneity equation modeling the treatment effect (λ) and a control equation.

3.1 R Code: Tomz and Weeks Reanalysis

First, the following model formula expresses a heterogeneous treatments model where the impact of alliances varies with the experimental conditions (regime, stakes, costs, and region). The `treat_group` indicator marks unique combinations of these experimental factors for the group-level random intercept. The control equation includes respondent demographics and foreign policy dispositions.

```
bf(
  force ~ lambda * alliance + controls,

  lambda ~ regime + stakes + costs + region_txt + (1 | treat_group),

  controls ~ white + male + hawk + intl + dem + rep + age + ed4,

  nl = TRUE
)
```

The second formula expresses a respondent heterogeneity model where the impact of alliances varies with demographic characteristics (party, news interest, race, and gender). The `het_group` indicator marks unique combinations of these demographic modifiers. The control equation includes experimental conditions and respondent age and education.

```
bf(
  force ~ lambda * alliance + controls,

  lambda ~ (dem + rep + high_newsint + white + male) + (1 | het_group),

  controls ~ regime + stakes + costs + africa + europe + asia + age + ed4,

  nl = TRUE
)
```

3.2 Priors: Tomz and Weeks Reanalysis

?? summarizes the prior distributions for both models. All priors are weakly informative relative to the scale of the data. The lambda coefficients receive a $N(0, 1)$ prior, while control

coefficients receive a tighter $N(0, 0.5)$ prior.

Parameter	Prior
λ coefficients	$N(0, 1)$
Control coefficients	$N(0, 1)$
σ (residual SD)	half- $N(0, 1)$
σ_θ (group SD)	half- $N(0, 1)$

Table 2. *Summary of priors in hierarchical models.*

3.3 *Who Responds to Alliances?*

To further explore the potential application of hierarchical models, this section examines how demographic factors modify the impact of alliances. I used party, news interest, race, and gender to modify the impact of alliances. I selected these variables because Tomz and Weeks examine party and news interest as potential modifiers, but gender (?) and race are also salient modifiers. I control for other experimental manipulations, age and education. Age and education are important group-level controls because they are group-level variables that are correlated across party, race and gender. Following TW's OLS analysis, I use a Gaussian likelihood, although the outcome is a binary variable.

The resulting hierarchical model thus encompasses four subgroup comparisons in TW's appendix while adding two more modifiers. All of the comparisons in the hierarchical model would require at least six pairwise interaction models. What TW present in four figures, I show with two.

To start, I present the two regressions from the hierarchical model. ?? plots how different variables shape either the impact of an alliance or support for intervention. One facet of this figure gives the coefficient estimates from the model of heterogeneity, and a second gives the control estimates.

High news interest is the strongest predictor of how alliances impact support for war. Respondents with high news interest respond to alliances by .15 more than others in this survey, and that relationship could be as small as .05 or as large as .22. This differs from TW's conclusion that respondents with high and low news interest respond in roughly the same way, likely because the hierarchical model accounts for other factors that are correlated with news interest.

Other demographic predictors do not differentiate responses to alliances as clearly. The impact of alliances may be somewhat greater for white and male respondents, but those differences are not clearly different from zero as the 95% credible intervals include zero and small negative values. Similarly, Republicans and Democrats are not all that different from independents in how they respond to alliances. This matches TW's conclusion that partisanship does not create substantial differences in alliance effects. The intercept in this equation is theoretically meaningful, as it captures the impact of alliances when all the modifiers are zero. For independent, non-white women with low news interest, alliances increase support for force by roughly .15.

The control equation suggests that high stakes, high costs, and partner democracy all increase support for intervention. Region is less consequential, as neither Asia, Eastern Europe nor Africa is clearly different from the reference category of Latin America. Because neither education nor age are equal to zero, the intercept is not directly interpretable.

While it is possible to construct a rough profile of who responds most to alliances using the estimates in ??, I provide a more precise summary in ??. This figure plots the median estimate in each group of respondents, after grouping respondents based on race, gender, news interest and partisanship. The median depends on the regression coefficients in ??, with additional noise from the partial pooling components. There are 24 unique estimates, all of which are clearly greater than zero. Alliances thus increase support for using force among all demo-

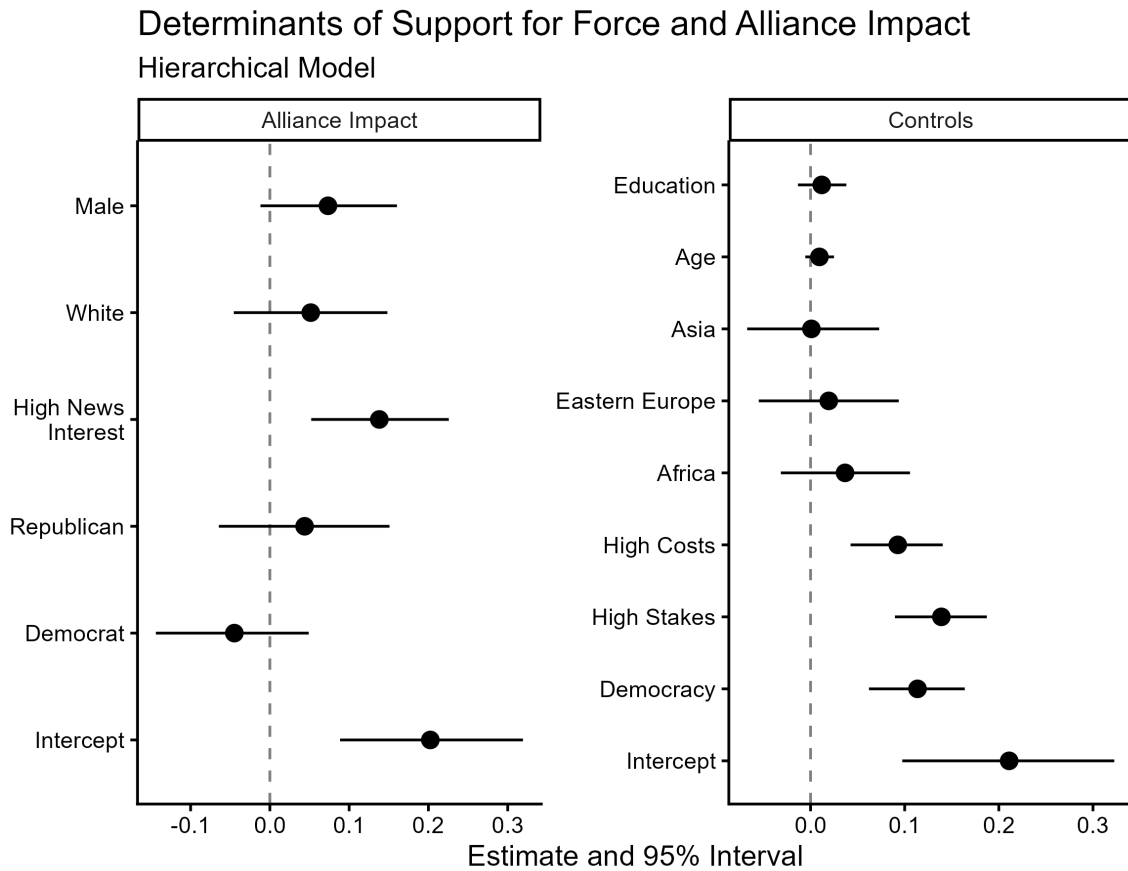


Figure 1. Regression coefficients from a hierarchical model of how an alliance shapes public support for military intervention. The left facet reports predictors of group alliance effects, and the right facet the control variables in the equation with the alliance treatment. Points give the posterior median and error bars the 95% credible interval.

Alliance Impact:
Party, Gender, Race, and News Interest

		Low News Interest		High News Interest		
Race	White	0.23	0.29	0.34	0.42	Democrat
	Non-White	0.13	0.22	0.3	0.38	
	White	0.25	0.32	0.39	0.45	Independent
	Non-White	0.2	0.29	0.35	0.41	
	White	0.3	0.37	0.44	0.5	Republican
	Non-White	0.25	0.32	0.38	0.46	
		Female	Male	Female	Male	Gender

Figure 2. *Estimated impact of alliance on support for military intervention in different subgroups of respondents. Text gives the exact posterior median estimate.*

graphic groups, but the magnitude of the effect varies.

Alliances most impact white Republican men with high interest in the news. In this group, the impact of alliances is .5, because this group was skeptical of intervention without an alliance rationale. The gap between high and low news interest in the other demographic groups is .15, which matches the coefficient in the heterogeneity equation.

The weakest impact of alliances appears for non-white, politically independent women with low news interest. Here, alliances increase support for force by 13%. White respondents respond by 8% more, Democrats by 4% more, and Republicans by 10%. The accumulation of these different effects create substantial gaps between the groups.

4 Bush and Prather Reanalysis

In the following, I again demonstrate how the model works by reanalyzing a study by ? (BP hereafter). This study examines how foreign meddling in elections impacts support for economic engagement with the meddler. One of their experiments examines how Russian or German engagement in the 2016 US election impacts mass support for trade and investment with those countries. I use a heterogeneous effects model to check their results and further explore their findings.

BP employ a 2x2x2 factorial experiment. This design randomizes whether a foreign country is interfering in the 2016 election, the country and their attitude towards Trump and whether the potential economic ties entail greater trade or investment in the United States. In one meddling treatment, Russia expresses support for Trump and in the other, Germany expresses opposition to Trump.

BP hypothesize that individuals will prefer economic engagement with states that support their candidate. They thus examine how the impact of side-taking varies with the direction of the endorsement, individual political preferences, and the economic issue. To do this, they use four tables and figures, and rely on eight t-tests of differences in means between groups of roughly 25 respondents.

I use the same nonlinear lambda specification as in the Tomz and Weeks reanalysis. The heterogeneous effects model estimates the impact of side-taking on economic engagement, with the treatment effect (λ) modeled as a function of potential modifiers. The heterogeneity equation includes a dummy indicator of intention to vote for Clinton, an indicator of whether the experiment deals with trade or investment, female gender, and high political engagement. I measure political engagement as a sum of political knowledge and interest, dichotomized at the median. Groups are defined by unique combinations of these four binary modifiers, producing 16 groups.

The outcome variable is a scale from one to four that measures support for greater economic engagement with the potential partner. To approximate BP's analyses, I use a normal outcome likelihood. The control equation includes college education and employment status.

4.1 R Code: Bush and Prather Reanalysis

The following model formula expresses the heterogeneous effects model:

```
bf(
  ipe_support ~ lambda * treat_germrus + controls,

  lambda ~ (w2_vote_hill + woman + high_pol_engage + invest_cond) +
    (1 | het_group),

  controls ~ college_educ + employ_dum + woman + high_pol_engage

  nl = TRUE
)
```

The priors follow the same structure as the Tomz and Weeks models, with $N(0, 1)$ on coefficients in the heterogeneity equation and $N(0, 0.5)$ on control coefficients.

These estimates corroborate BP's findings. Clinton voters are more likely to support economic engagement with Germany when Germany has opposed Donald Trump. This has the largest impact on attitudes towards trade among highly engaged respondents. Low engagement Clinton voters are less responsive.

Trump voters are less responsive to German opposition to Trump. The effects are mostly null, and only for engaged voters in the investment condition does the endorsement have a clearly non-zero effect. This positive effect may reflect residual opposition to Trump in some sections of the Republican party.

References

- Barnhart, Joslyn N, Robert F Trager, Elizabeth N Saunders and Allan Dafoe. 2020. "The Suffragist Peace." *International Organization* 74(4):633–670.
- Bush, Sarah Sunn and Lauren Prather. 2020. "Foreign Meddling and Mass Attitudes Toward International Economic Engagement." *International Organization* 74(2):584–609.
- Tomz, Michael and Jessica L.P. Weeks. 2021. "Military Alliances and Public Support for War." *International Studies Quarterly* 65(3):811–824.

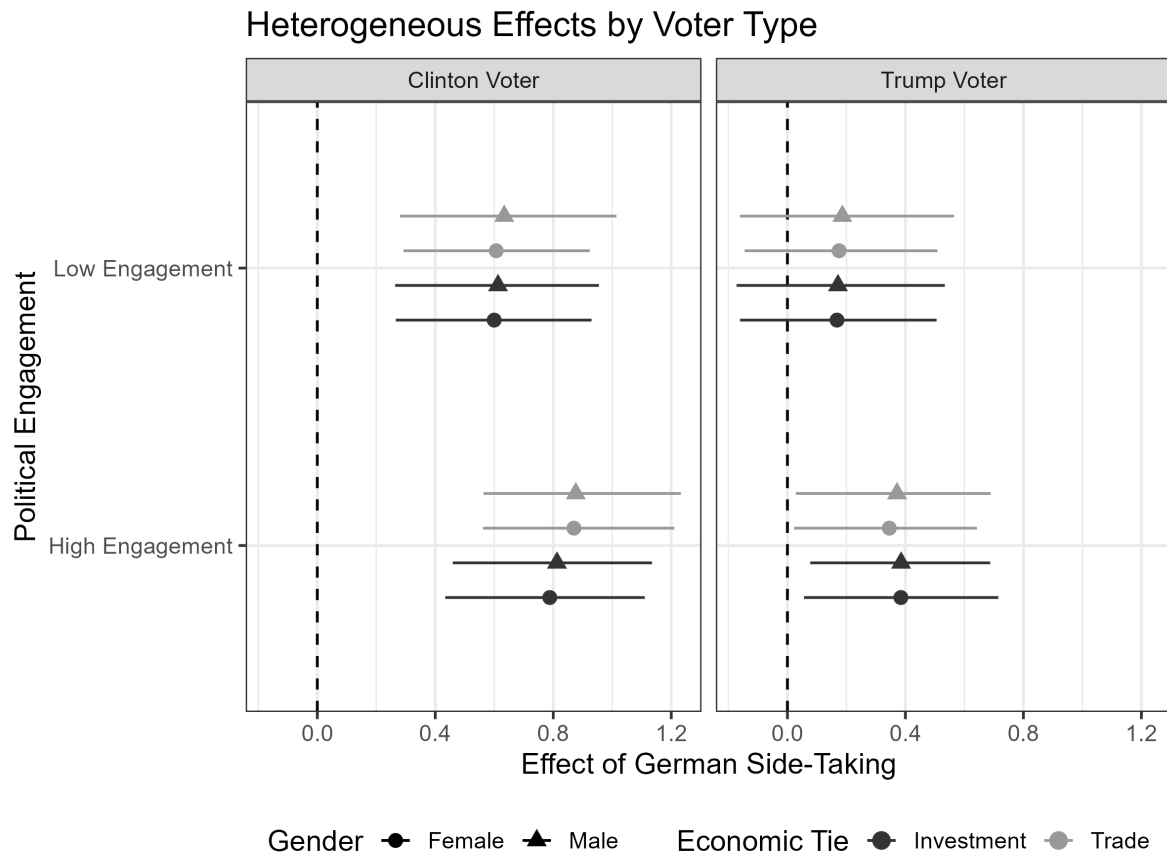


Figure 3. *Heterogeneous effects of German side-taking in US elections on support for economic engagement with Germany. Each estimate reflects a treated group with a unique combination of other treatments and demographic characteristics.*